

HEALTH-RING — Production Datasheet

EoS Health · Models EOS-HR-001 (Base) / EOS-HR-002 (Ultra) · Revision 1.0

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1. Product Overview

HEALTH-RING is a titanium finger ring health monitor available in two tiers. The Base tier delivers ECG, 3-wavelength PPG, skin temperature, and motion tracking in the world's thinnest ECG ring at 2.0 mm. The Ultra tier adds 5-wavelength NIR spectroscopy for non-invasive HbA1c estimation, cuffless blood pressure via piezoelectric pulse transit time, and an on-device AI co-processor — capabilities no other ring-form-factor device offers.

2. Tier Comparison

Specification	Base (EOS-HR-001)	Ultra (EOS-HR-002)
Ring width	2.0 mm	2.8 mm
Ring profile	2.5 mm	3.2 mm
Weight	3.2 g	4.8 g
Battery	60 mAh	170 mAh
Battery life	4 days	7 days
Charging	NFC inductive (13.56 MHz)	NFC inductive (13.56 MHz)
MCU	nRF52833	nRF52840 + MAX32666 AI
PPG wavelengths	3 (660/850/940 nm)	5 (660/730/850/940/1300 nm)
ECG electrodes	ZPIES flush gold	DAEA dual-arc Pt-Ir
HbA1c	—	✓ ±0.23% ARMS
Cuffless BP	—	✓ ±5 mmHg
AFib detection	✓	✓ AUC 0.998
SpO ₂	✓ ARMS 0.44%	✓ ARMS 0.44%
Retail price	\$199	\$399

3. Electrical Specifications

Base (EOS-HR-001)

Parameter	Min	Typ	Max	Unit
Battery capacity	—	60	—	mAh
Active current	—	6.2	9	mA
Deep sleep current	—	3.1	5	μA
NFC charging power	—	0.5	1.0	W
Charging time	—	90	120	min

Ultra (EOS-HR-002)

Parameter	Min	Typ	Max	Unit
Battery capacity	—	170	—	mAh
Active current	—	9.8	14	mA
Deep sleep current	—	3.8	6	μA
NFC charging power	—	1.0	2.0	W
Charging time	—	90	120	min

4. Sensor Suite

Sensor	IC	Tier	Measurement	Accuracy
ECG (ZPIES)	ADS1291	Base	Single-lead ECG, HR, HRV, AFib	SNR $\geq$ 58 dB
ECG (DAEA)	ADS1292R	Ultra	Dual-arc ECG, Lead I+II	SNR $\geq$ 63 dB
PPG 3 λ	MAX30102	Base	SpO ₂ , HR, HRV	ARMS $\leq$ 0.44%
PPG 5 λ (MSHE)	MAX86176	Ultra	SpO ₂ , HbA1c, HbO ₂ , MetHb	ARMS $\leq$ 0.44%
Piezo BP (PPTT)	LDT0-028K	Ultra	Cuffless systolic/diastolic BP	$\pm$ 5 mmHg
Accelerometer	LIS2DH12	Both	3-axis motion, sleep staging	$\pm$ 0.05 g
Temperature	MCP9808	Both	Skin temperature, circadian	$\pm$ 0.25°C
KEHS harvester	LTC3588-1	Base	Kinetic energy $\rightarrow$ +18% battery	—

5. Mechanical Specifications

Parameter	Base	Ultra
Ring width	2.0 mm	2.8 mm
Ring profile height	2.5 mm	3.2 mm
Inner diameter options	14–23 mm (sizes 3–13 US)	14–23 mm (sizes 3–13 US)
Material	Grade 5 Titanium (Ti-6Al-4V)	Grade 5 Titanium (Ti-6Al-4V)
Finish	Polished / Matte / PVD Black	Polished / Matte / PVD Black / Rose Gold
IP rating	IP68 (200 m, 2 hr)	IP68 (200 m, 2 hr)
PCB type	Flexible PCB (FPCB), 4-layer	Flexible PCB (FPCB), 6-layer
Electrode material	24K gold-on-anodized-Ti (ZPIES)	Platinum-iridium arc (DAEA)

6. Processor and Memory

Component	Base	Ultra
MCU	nRF52833 (Cortex-M4F, 512 KB Flash)	nRF52840 (Cortex-M4F, 1 MB Flash)
AI co-processor	—	MAX32666 (dual M4F, TFLite)
NVM	W25Q16JV (2 MB)	W25Q32JV (4 MB)
Secure element	ATECC608B	ATECC608B

7. BOM Summary (Production, 1k units)

Category	Base Cost	Ultra Cost
MCU + memory	\$4.20	\$8.80
Sensors	\$7.40	\$14.60
Power (PMIC + battery + NFC coil)	\$5.80	\$8.20
Flexible PCB	\$6.50	\$9.80
Titanium ring body + machining	\$8.20	\$11.40
Assembly + test	\$3.50	\$4.50
Charging dock	\$4.80	\$6.20
Packaging	\$2.10	\$3.20
Total BOM	\$42.50	\$66.70
Target retail	\$199	\$399

8. Charging Dock Specifications

Parameter	Value
Charging standard	NFC inductive, 13.56 MHz
Input	USB-C 5V/1A
Charging current	100–500 mA (adaptive)
Dock material	Anodized aluminum
Dimensions	40 × 40 × 12 mm
Included in box	Yes (both tiers)